

SYDNEY TECHNICAL HIGH SCHOOL

MATHEMATICS EXTENSION I

HSC ASSESSMENT TASK 1 - 2008

Students' name: _____ Teacher's name: _____

Q1	Q2	Q3	Q4	Q5	Q6	Total
						/ 50

Time allowed: 70 minutes

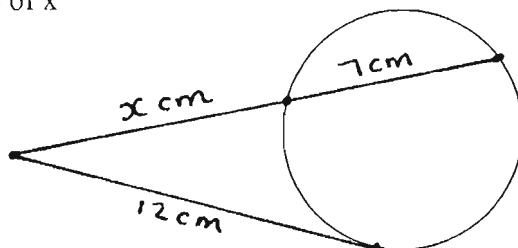
- Attempt all questions
- Show all necessary working.
- Marks may not be awarded for insufficient working or poorly set out solutions.

START EACH QUESTION ON A NEW PAGE

Question 1 (8 marks)

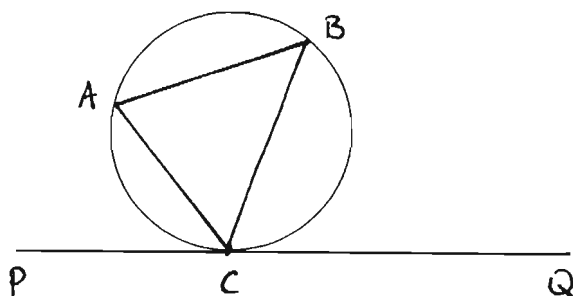
a) Find the value of x

2



b) Explain why angle PCA equals angle CBA.

1



- c) i. Insert 6 terms between -10 and 165 so that the series is arithmetic. **2**

- ii. If this pattern continues find the sum of the first 20 terms. 2

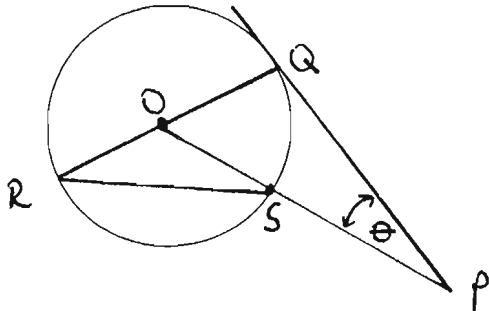
d) Evaluate $\sum_{n=3}^7 n^3$ 1

Question 2 (start a new page) (8 marks)

- a) Given $S_n = 17n - 3n^2$, find an expression for the n th term. 3

- b) Which term of the series, $100 + 20 + 4 + \dots$,
is the first term with a value of less than 10^{-4} 3

- c) Given QP is a tangent to a the circle centre O, 2
- Find, the size of angle ORS in terms of θ . Give reasons.



Question 3 (start a new page) (8 marks)

a) Given $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$ 3

Show that $\sum_{n=1}^{K+1} \frac{1}{n(n+1)} = \frac{k+1}{k+2}$

- b) $P(2ap, ap^2)$ is any point on the parabola $x^2 = 4ay$.
The line l passes through the focus, S, and is parallel to the tangent at P.

- i. Find the equation of the line l 2
- ii. The line l intersects the x – axis at Q.
Write down the coordinates of Q. 1
- iii. Find the equation of the locus of the midpoint of QS. 2

Question 4 (start a new page) (9 marks)

- a) Consider the series, $x + 4x^2 + 16x^3 + \dots$

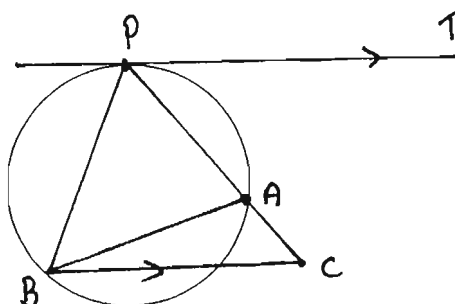
- i. For what value of x does the series have a limiting sum 1
- ii. Find the value of x given $S_{\infty} = \frac{3}{2}$ 2

- b) Tom borrows \$250 000 to buy a unit in Cronulla.
The rates are 6% p.a. monthly reducible interest and equal monthly repayments of \$1900 are payable at the end of each month.

- i. How much does Tom owe immediately after the second repayment. 1
- ii. Show that after n months Tom owes
\$ $(380\,000 - 130\,000(1.005)^n)$ 2
- iii. Find the balance owing at the end of 5 years 1
- iv. If after 5 years the interest rate increases to 7.2% p.a.,
find the minimum monthly repayment needed to repay
the loan in a further 20 years. 2

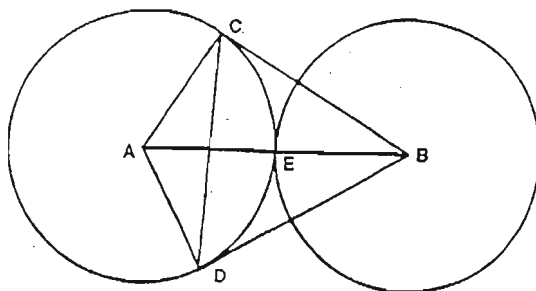
Question 5 (start a new page) (8 marks)

- a) Prove by mathematical induction that $13 \times 6^n + 2$ is divisible by 5 for every positive integer n . 4
- b) In the diagram A, P and B are points on the circle. The chord PA is produced to C and BC is parallel to the tangent at P.



- i. Show that Angle PBA = Angle PCB 2
- ii. Deduce that $PB^2 = PA \times PC$ 2

Question 6 (start a new page) (9 marks)



Two circles of equal radius and with centres at A and B respectively touch each other externally at E. BC and BD are tangents from B to the circle with centre A.

- i. Copy the diagram 1
- ii. Show that BCAD is a cyclic quadrilateral. 2
- iii. Show that E is the centre of the circle which passes through B, C, A and D. 2
- iv. Show that angle CBA = angle DBA = 30° 2
- v. Show that triangle BCD is equilateral 2

2008 HSC task I Extension I.

Question 1.

a)

$$12^2 = x(x+7) \quad \checkmark$$

$$144 = x^2 + 7x$$

$$0 = x^2 + 7x - 144$$

$$0 = (x+16)(x-9)$$

$$x > 0 \therefore x = 9. \quad \checkmark$$

b) The angle between a tangent and a chord is equal to the angle in the alternate segment. $\checkmark$

c)

i. $-10, -, -, -, -, -, -, 165$ $\frac{AP}{\checkmark}$

$$165 = -10 + 7d \quad \checkmark$$

$$175 = 7d$$

$$d = 25 \quad \checkmark$$

$$\therefore 15, 40, 65, 90, 115, 140$$

ii. $S_{20} = \frac{20}{2}(-20 + 19 \times 25) = 4550 \quad \checkmark$

d) $\sum = 3^3 + 4^3 + 5^3 + 6^3 + 7^3 = 775. \quad \checkmark$

Question 2.

a) $T_n = S_n - S_{n-1} \quad \checkmark$
 $= 17n - 3n^2 - [17(n-1) - 3(n-1)^2]$
 $= 17n - 3n^2 - [17n - 17 - 3n^2 + 6n - 3]$
 $= 17n - 3n^2 - [-3n^2 + 17n + 6n - 20]$
 $= 20 - 6n \quad \checkmark$

b) $100 + 20 + 4 + \dots \quad a = 100$

$$r = \frac{1}{5}$$

$$ar^{n-1} < 10^{-4} \quad \checkmark \quad T_n = 10^{-4}$$

$$100 \times \left(\frac{1}{5}\right)^{n-1} < 10^{-4}$$

$$0.2^{n-1} < 0.000001 \quad \checkmark$$

LOGS

$$(n-1) \log 0.2 < \log 10^{-6}$$

$$n-1 > 8.584 \dots$$

$$n > 9.584 \dots$$

$$n = 10$$

(OR) Trial & error

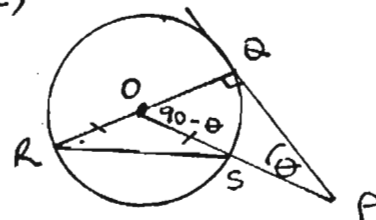
$$0.2^9 = 5.12 \times 10^{-7}$$

$$0.2^8 = 2.56 \times 10^{-6}$$

$$\therefore n-1 = 9$$

$$n = 10$$

c)



$\angle OQP = 90^\circ$ (radius to tangent is 90°)

$\angle AOS = 90 - \theta$ (angle sum $\triangle OQP$)

$\angle ROS = 90 + \theta$ (angles on a straight line)

$RO = OS$ radii

$\therefore \angle ORS = \angle OSR$ (equal angles opp equal sides)

$$\angle ORS = \frac{90 - \theta}{2} \quad \checkmark$$

Question 3

a) $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$

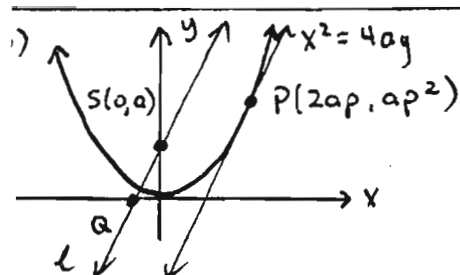
$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \quad \checkmark$$

$$= \frac{k(k+2) + 1}{(k+1)(k+2)} \quad \checkmark$$

$$= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$$

$$= \frac{(k+1)^2}{(k+1)(k+2)}$$

$$= \frac{k+1}{k+2} \quad \text{as required} \quad \checkmark$$



$$1. y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a} \text{ at } x = 2ap$$

$$m_T = p \text{ as } l \text{ is } // \text{ to tangent}$$

$$m_2 = p \text{ thru } (0, a) \quad \checkmark$$

$$\therefore l: y - a = p(x - 0)$$

$$y = px + a \quad \checkmark$$

$$II. Q: y = 0 \quad 0 = px + a$$

$$-\frac{a}{p} = x$$

$$\therefore (-a/p, 0)$$

$$III. \text{Midpt} \left[\frac{-a/p + 0}{2}, \frac{0 + a}{2} \right] \quad \checkmark$$

$$\left[-a/2p, a/2 \right]$$

$$\text{Locus of midpt} \Rightarrow y = \frac{1}{2}a \quad \checkmark$$

Question 4.

$$a) x + 4x^2 + 16x^3 + \dots$$

$$1. -1 < r < 1$$

$$-1 < 4x < 1$$

$$-1/4 < x < 1/4 \quad \checkmark$$

$$II. \frac{x}{1-4x} = \frac{3}{2} \quad \checkmark$$

$$2x = 3 - 12x$$

$$14x = 3$$

$$x = \frac{3}{14} \quad \checkmark$$

$$b) 6\% p.a = 0.005 \text{ a month}$$

$$I. A_1 = 250\,000(1.005) - 1900$$

$$A_2 = 250\,000(1.005)^2 - 1900(1.005) - 1900$$

$$II. A_3 = 250\,000(1.005)^3 - 1900[1.005^2 + 1.005 + 1] \quad \checkmark$$

$$A_n = 250\,000(1.005)^n - 1900[1.005^{n-1} + 1.005^{n-2} + \dots + 1] \quad \checkmark$$

$$A_n = 250\,000(1.005)^n - 1900 \left[\frac{1(1.005^n - 1)}{0.005} \right] \quad \checkmark$$

$$= 250\,000(1.005)^n - 380\,000(1.005^n - 1)$$

$$= 380\,000 - 130\,000(1.005)^n$$

$$III. n = 5 \times 12 = 60$$

$$A_{60} = 380\,000 - 130\,000(1.005)^{60}$$

$$= 204649.48 \quad \checkmark$$

IV.

$$0 = 204649.48(1.006)^{240} - m \left[\frac{1.006^{240} - 1}{0.006} \right] \quad \checkmark$$

$$m \left[\frac{1.006^{240} - 1}{0.006} \right] = 204649.48(1.006)^{240}$$

$$m = \$1611.32$$

Question 5

a) Test $n=1$

$$13 \times 6^1 + 2 = 80 \\ = 5 \times 16 \therefore \text{divisible by } 5$$

Assume true for $n=k$

ie $13 \times 6^k + 2 = 5M$ where M is a +ve integer

Prove true for $n=k+1$

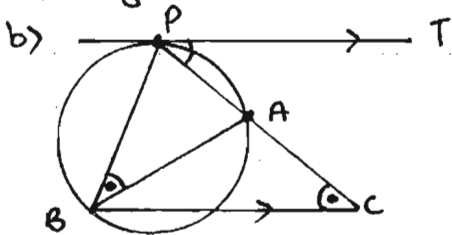
$$13 \times 6^{k+1} + 2 \\ = 6 \times 13 \times 6^k + 2 \\ = 6[5M - 2] + 2 \\ = 30M - 12 + 2 \\ = 30M - 10 \\ = 5[6M - 2]$$

which is $\div$ by 5 $\therefore$ true $n=k+1$.

If true $n=k$ also true $n=k+1$

As true $n=1$ also true $n=2, 3, 4, \dots$

Hence by M.I true all positive integer n .



angle between a tangent

- i. $\angle PBA = \angle TPC$ (and a chord equals angle in alternate segment)
- $\therefore \angle BCP = \angle TPC$ (alternate angles $PT \parallel BC$)
- $\therefore \angle PBA = \angle PCB$.

ii. In $\triangle PBA$ and $\triangle PCB$

$\angle P$ is common

$\angle PBA = \angle PCB$ (part i)

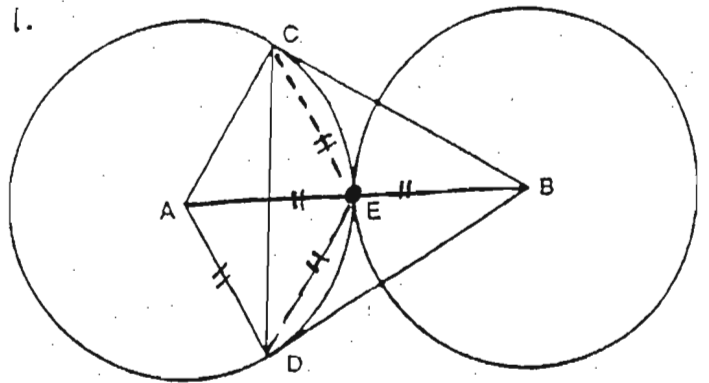
$\therefore \triangle PBA \sim \triangle PCB$ (equiangular)

$$\therefore \frac{PB}{PC} = \frac{PA}{PB} \quad (\text{ratio of corresp sides})$$

$$\therefore PB^2 = PA \times PC$$

Question 6

i.



- ii. $\angle ACB = 90^\circ$ (radius to tangent)
 $\angle ADB = 90^\circ$ (radius to tangent)
 Now $\angle ACB + \angle ADB = 180^\circ$
 and BCAD is cyclic (opposite angles supplementary)

iii. As $\angle ACB = 90^\circ$

AB is a diameter (angle in semi circle)

As $AE = EB$

radii of equal circles

E is the midpoint of AB

and the centre of circle B, C, A, D.

iv. As E is the centre of circle

$$\therefore CE = EB = ED$$

$$AE = AD \text{ radii centre A.}$$

$$\therefore AE = ED = AD$$

and $\angle DAE = 60^\circ$

$$\angle ADB = 90^\circ \text{ (radii to tangent)}$$

$$\angle DBA = 30^\circ \text{ (angle sum } \triangle ABD)$$

Likewise $\angle ABC = 30^\circ$

$$\therefore \angle DBA = \angle ABC = 30^\circ$$

v.

$CB = DB$ tangents from external point equal

$$\therefore \angle BCD = \angle BDC \text{ (equal angles opposite equal sides)}$$

$$\angle CBD = 60^\circ \text{ (part iv)}$$

$$\therefore \angle BCD = \angle BDC = 60^\circ \text{ (angle sum } \triangle BCD)$$

$\therefore \triangle BCD$ is equilateral.